Juju

Juju is an open source configuration management and orchestration management tool. It enables applications to be deployed, integrated and scaled on various types of cloud platforms faster and more efficiently. It allows users to export and import application architectures and reproduce the same environment at different phases on cloud platforms such as Joyent, Amazon Web Services, Windows Azure, HP Cloud and IBM.

It is used to configure, deploy, scale, manage, monitor, diagnose, notify and efficiently maintain (through the use of best practices) Charms (applications as services) in any deployment model, such as public, private and hybrid clouds —either from a powerful GUI or the command-line. It also supports customisation of Charms using Chef, Puppet or any other CM tool.

The main mechanism behind Juju is known as Charms that can be written in any programming language, whose execution is supported via the command line. They are a collection of YAML configuration files.

Clients are available for Ubuntu, Windows and Mac operating systems. Once you install the client, environments can be bootstrapped on various cloud platforms such as Windows Azure, HP Cloud, Joyent, Amazon Web Services and IBM.

Juju supports the GUI as well as the command line. It has a more feature-rich command line in comparison to what can be achieved via the GUI, which supports the drag and drop feature to create a software stack. Juju is limited to Ubuntu on the server side. Its Python version has a community patch to support CentOS.

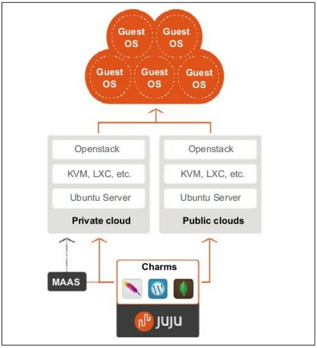

Figure 4 : Juju deployment

Rudder

Rudder is an open source CM tool for managing IT infrastructures. It is written in Scala and works on top of the CFEngine. Rudder provides features such as host inventory, a Web interface, reusable policies, a policy editor, generation of per-host policies, API, change requests, etc.

Rudder's asset management function identifies nodes and their characteristics that can be useful to perform configuration management. CM applies rules on nodes. Rudder uses the asset management function to identify nodes for configuration management.

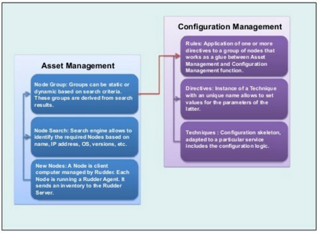

Figure 5: Rudder

Ansible

Ansible is an open source platform for CM, orchestration and deployment of compute resources. It manages resources with the use of SSH (Paramiko, a Python SSH2 implementation, or standard SSH). Ansible supports and is available for CentOS and Red Hat Enterprise Linux, and it is also available as a commercial product by Ansible Inc. It is built on the popular Python language. It is possible to install Ansible by using the Git repository clone of a master server. Ansible is agentless. Its design goals are consistent, secure, minimal in nature and highly reliable, and it is easy to learn.

Its main features are:

- Resources are added to the Ansible configuration.
- SSH authorised keys or sudo credentials (root access is not needed) are needed for each managed compute resource based on the user.
- Ansible's master server communicates with the compute resources using SSH and performs all the necessary tasks.
- Ansible deploys modules to compute resources.
- It monitors quantities such as CPU resources.

Playbooks are configuration files in Ansible, which use YAML syntax. Ansible has a collection of modules to manage resources on various cloud platforms such as Amazon EC2 and OpenStack.

Ansible supports deployments on various virtualisation platforms as well as public and private cloud environments